Middle School Chemistry Syllabus

Course Title

Middle School Chemistry

Grade Level

Grades 6–8

Course Length

Full academic year

Course Description

Middle School Chemistry is a teacher-directed, whole-class introduction to the fundamental ideas of chemistry.

All students will progress through the same topics together according to the course schedule. Instruction will combine direct teaching, visual learning, class discussion, demonstrations, guided practice, laboratory investigations, and assessments.

Students will use **Barron's Visual Learning: Chemistry** as the primary text, supplemented by teacher-created materials, selected ACS Middle School Chemistry activities, simulations, and laboratory investigations.

The goal of the course is to develop a strong conceptual understanding of matter and chemical change while preparing students for high-school chemistry.

Primary Resource

Barron's Visual Learning: Chemistry

Additional resources may include:

- Teacher-created notes and worksheets
- ACS Middle School Chemistry
- PhET simulations
- Laboratory investigations
- Teacher demonstrations
- Videos and visual resources

Course Format

This is a **teacher-paced course**.

Students will:

- Study the same unit at the same time.
- Complete common assignments and activities.
- Participate in labs as a class.
- Follow established assignment deadlines.
- Take quizzes and tests on scheduled dates.
- Review and practice material together before major assessments.

Students who are absent or fall behind are responsible for completing missed work and returning to the class schedule as quickly as possible.

Course Objectives

By the end of the course, students should be able to:

- Describe matter using physical and chemical properties.
- Explain solids, liquids, and gases using particle models.
- Distinguish between physical and chemical changes.
- Calculate and interpret density.
- Describe basic atomic structure.
- Identify protons, neutrons, and electrons.
- Explain atomic number, mass number, and isotopes.
- Read basic information from the periodic table.
- Describe groups, periods, and common periodic patterns.
- Explain valence electrons conceptually.
- Describe the formation of ions.
- Distinguish between ionic and covalent bonding.
- Interpret basic chemical formulas.
- Describe molecules, molecular shape, and polarity conceptually.
- Identify evidence of chemical reactions.
- Explain conservation of matter.
- Describe factors affecting reaction rates.
- Explain solutions and solubility.
- Distinguish among acids, bases, and neutral substances.
- Interpret the pH scale.
- Describe basic gas behavior.
- Distinguish between endothermic and exothermic processes.
- Conduct scientific investigations safely.
- Record observations and measurements accurately.
- Use scientific evidence to support conclusions.

Course Units

Quarter 1 — Matter and Atomic Foundations

Unit 1 — Matter and Measurement

- Matter
- Physical properties
- Chemical properties
- Solids, liquids, and gases
- Mass
- Volume
- Density
- Physical and chemical changes

Unit 2 — Atomic Structure

- Atoms
- Protons
- Neutrons
- Electrons
- Atomic number
- Mass number
- Isotopes

Quarter 2 — Periodic Table and Bonding

Unit 3 — The Periodic Table

- Elements
- Periods
- Groups
- Metals
- Nonmetals
- Valence electrons
- Periodic patterns

Unit 4 — Chemical Bonding

- Ions
- Ionic bonding
- Covalent bonding
- Molecules
- Chemical formulas
- Molecular structure
- Polarity

Quarter 3 — Reactions and Solutions

Unit 5 — Chemical Reactions

- Reactants and products
- Evidence of chemical change
- Conservation of matter
- Reaction energy
- Reaction rates
- Catalysts

Unit 6 — Solutions

- Solutes
- Solvents
- Dissolving
- Solubility
- Concentration
- Water and polarity

Quarter 4 — Acids, Bases, Gases, and Energy

Unit 7 — Acids and Bases

- Acids
- Bases
- Indicators
- pH
- Neutralization

Unit 8 — Gases and Energy

- Gas particles
- Pressure
- Temperature
- Volume
- Endothermic processes
- Exothermic processes

Unit 9 — Chemistry Review and Integration

Students will review and apply concepts from throughout the year through cumulative activities, investigations, and final assessments.

Typical Class Structure

A typical class period may include:

1. Bellwork or review

2. Teacher-led lesson
3. Barron’s reading or visual reference
4. Guided examples or class discussion
5. Individual or partner practice
6. Demonstration, laboratory activity, or simulation when appropriate
7. Assignment or exit check

The class will generally move forward together after adequate instruction and review.

Laboratory Program

Laboratory investigations will be scheduled throughout the year and completed as a class.

Possible investigations include:

- Density of unknown materials
- Separation of mixtures
- Solubility
- Conservation of mass
- Evidence of chemical reactions
- Reaction rates
- Acids and bases
- Energy changes

Some topics may be taught through demonstrations or simulations when those methods are safer, clearer, or more appropriate than a physical experiment.

Assignments

Students may receive:

- Reading assignments
- Vocabulary work
- Guided notes
- Worksheets
- Practice questions
- Lab sheets
- Short written explanations
- Projects
- Review assignments

Assignments will have common due dates.

Late work will be handled according to school and teacher policy.

Assessment

Students may be evaluated using:

- Classwork and homework
- Quizzes
- Unit tests
- Lab participation
- Lab reports or lab sheets
- Projects
- Quarter assessments
- Final examination or investigation

Grading Policies

Quarterly grades are derived from two distinct components: **Academic Performance** (evaluated assignments) and **Participation** (teacher-assessed engagement and preparation). While most courses utilize an equal 50/50 weighting, specific courses may be designated as Skill-Heavy (60/40) or Participation-Heavy (40/60) at the teacher's discretion.

The Academic component is calculated by converting each assignment into a percentage and a corresponding letter grade. This figure represents a combination of the overall quarterly trend (60%) and the average of the five most recent assignments (40%). This structure ensures that student improvement in the latter half of the quarter significantly influences the final grade.

Every assessment carries equal weight within the grading system. Small quizzes and major projects exert the same influence on the final average, emphasizing the importance of consistent performance over any individual result.

Grades are recorded using a letter-based system where percentages are snapped to established midpoints. Consequently, numerical differences within the same letter-band (such as 84% and 86%) result in the same grade. Students should focus on achieving higher letter-bands rather than individual points.

Missing assignments will be recorded as a zero after the designated grace period. Until that time, the displayed grade only reflects completed work, which may not accurately represent a student's current academic standing.

Grading Scale: A+ 97–100 · A 93–96 · A– 90–92 · B+ 87–89 · B 83–86 · B– 80–82 · C+ 77–79 · C 73–76 · C– 70–72 · D+ 67–69 · D 63–66 · D– 60–62 · F below 60

The final year grade is weighted: Q1 20%, Q2 25%, Q3 25%, and Q4 30%. Cumulative GPA is calculated on a 4.3 scale, with an A+ representing a 4.3.

Tests and Quizzes

Quizzes will generally assess recent material.

Unit tests will assess the major concepts taught during each unit and may include selected material from previous units.

Students should expect chemistry knowledge to build throughout the year.

Review will normally occur as a class before major assessments.

Laboratory Expectations

Students must:

- Follow all teacher instructions.
- Wear required safety equipment.
- Handle chemicals and equipment only as directed.
- Maintain a clean work area.
- Record observations accurately.
- Properly dispose of materials.
- Immediately report spills or breakage.
- Never taste laboratory substances.
- Never perform unauthorized experiments.

Unsafe behavior may result in removal from the laboratory activity.

Student Materials

Students should bring:

- Barron’s Visual Learning: Chemistry
- Chemistry notebook or binder
- Pencil
- Calculator when requested
- School-issued device when requested

Academic Expectations

Students are expected to:

- Arrive prepared and on time.
- Participate in class instruction and discussion.
- Complete assignments by their deadlines.
- Keep an organized chemistry notebook.
- Ask questions when clarification is needed.
- Follow laboratory safety expectations.

- Work respectfully with classmates.
- Complete their own assessed work.

Final Course Goal

By the end of Middle School Chemistry, students should understand chemistry as a connected progression:

matter → particles → atoms → bonding → molecules → reactions

Students should be prepared to enter high-school chemistry with a strong conceptual foundation and familiarity with scientific investigation.